

Theofilus Hobba Pramono

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🔗 theohobba.github.io in Theofilus Hobba Pramono

Profile

An astronomy graduate with experience in light curve modeling and science communication research. Interested in detection and characterization of circumstellar and circumplanetary discs, light curve modeling, planet formation research, as well as cultural astronomy research.

Education

- M.Sc Leiden University**, Astronomy and Science Communication and Society (SCS) Aug 2022 – Jul 2024
- GPA: 7.93/10.00
- B.Sc Bandung Institute of Technology**, Astronomy Aug 2017 – Jul 2021
- GPA: 3.52/4.00

Research Experience

Societal Impacts of Timau National Observatory, Indonesia: an Exploratory Study, supervised by **Dr. Pedro Russo** Jan 2024 - Jul 2024

Research internship project to complete the science communication component of Leiden University's Astronomy and SCS master's program.

- Identified existing and potential societal opportunities and issues of Timau National Observatory by assessing stakeholder accounts using the qualitative content analysis approach.
- Developed a strategic communication plan based on research results and strategic plans from relevant research institutes.

Follow Up of ASASSN-21js Light Curve, supervised by **Dr. Matthew Kenworthy** Oct 2022 - Jul 2023

Master's thesis project to complete the astronomy component of Leiden University's Astronomy and SCS master's program.

- Built transiting ring system models using the `exorings3` Python package and MCMC to fit the available light curve data of ASASSN-21js' asymmetric and on-going dimming event.
- Explored probable ring system geometries and tried to predict the remaining light curve trend and end date by sampling various ring geometries using MCMC.

Exoplanet Search among Southern Galactic Open Clusters, mentored by **Denny Mandey, M.Si.** Mar 2022 - Jul 2022

Research program at Bosscha Observatory, Indonesia.

- Prepared and conducted time series observations at NGC 6416 and NGC 5460 using Bosscha Observatory's 27.9 cm reflector telescope.

Archaeoastronomy Study of Sukuh and Cetho Temples in Central Java, supervised by **Dr. Aprilia** Sep 2020 - Jun 2021

Bachelor's thesis project at Bandung Institute of Technology, Indonesia.

- Modeled the alignments between two Javanese Hindu temples and positions of the Sun, Moon, and culturally significant bright stars around the 15th century CE.

Publication

Variability Survey of Southern Open Clusters NGC 6134, NGC 6416, Collinder 394, and Alessi 10

Feb 2026

Mandey, D., Ramadhan, D. G., Yusuf, M., Perhati, T., Ramadhan, S., **Pramono, T. H.**, Anin, C. J., Premadi, P. W., Jaelani, A. T., Jatmiko, A. T. P., & Arifyanto, M. I.

<https://doi.org/10.5614/j.math.fund.sci.2026.57.3.1>

ASASSN-21js: A Multi-year Transit of a Ringed Disc

Aug 2024

T. H. Pramono, M. A. Kenworthy, R. van Boekel

[10.1051/0004-6361/202450288](https://doi.org/10.1051/0004-6361/202450288)

Conferences

Southeast Asia Astronomy Network Meeting (SEAN), Bandung Institute of Technology, Indonesia

13-15 Oct 2025

- Presented a talk in English titled "Exploring the Societal Impacts of Timau National Observatory, Indonesia".

Observational Astronomy Seminar and Discussion, Bandung Institute of Technology, Indonesia

9-11 Apr 2025

- Presented a talk in Indonesian titled "ASASSN-21js and Analysis Opportunities on Light Curves featuring Unique Dimming Events".

Funding and Awards

Indonesian Endowment Fund for Education Scholarship (LPDP)

Jul 2022 - Jul 2024

Awarded a fully-funded scholarship from the Indonesian government to pursue a two years master's degree at Leiden University.

Dean's List for Academic Performance of 1st and 2nd Semester of 2019/2020

Bandung, Indonesia
2019-2020

Earned a minimum GPA of 3.50/4.00 in a particular semester with B being the minimum grade as an undergraduate student at the Faculty of Mathematics and Natural Science of Bandung Institute of Technology.

Work Experience

Stargazing Session Guide, Deepsky Villa

Karimunjawa, Indonesia
Aug 2026 - Present

- Managed session reservations and developed session flow and materials.
- Guided guests in observing planets, stars, clusters, and nebulae using a telescope and explained the science behind these objects.
- Helped with telescope maintenance.

Freelance Astronomy Content Writer and Editor, Ditara.id

Indonesia (remote)
Oct 2022 - Sep 2023

- Wrote popular astronomy contents on predetermined topics to publish as Instagram feeds and reels.
- Edited content drafts based on guidelines to maintain content quality.

Public Outreach Staff, Bosscha Observatory

Lembang, Indonesia
Apr 2019 - Jul 2022

- Operated telescopes, organized crowd flow, and engaged with visitors during day and night time public visits.

Teaching Assistant, Bandung Institute of Technology

Bandung, Indonesia
Sep 2020 - Dec 2020

- Tutored undergraduate students enrolled in the Astrophysics I class on a weekly basis and graded class assignments throughout the semester.

Volunteer Experience

Science Communication Student Conference Volunteer, Leiden University

Leiden, The Netherlands

Dec 2023 - May 2024

- Worked in a team to select the main theme and topics, design visual graphic assets, develop activities, and arrange the timeline for the national conference for science communication students in the Netherlands.

Pale Blue Dot Carolina Link (PBD C-Link) Team Member, Leiden University

Leiden, The Netherlands

Oct 2023 - Feb 2024

- Involved in a team led by Prof. Dr. George K. Miley to organize an international online meeting promoting astronomy and cultural values for elementary students in five schools across Ireland, The Netherlands, Ethiopia, Armenia, and Nigeria.

Head of Science and Project Department, Astronomy Undergraduate Student Association of Bandung Institute of Technology (Himastron ITB)

Bandung, Indonesia

Jan 2020 - Jan 2021

- Coordinated three divisions to provide Himastron ITB members with opportunities to sharpen their astronomy research skills and practice public outreach.

Referees

Dr. Matthew A. Kenworthy

Associate Professor at Leiden University, The Netherlands

Email: kenworthy@strw.leidenuniv.nl

Phone: +31 (0) 71 527 8455

Dr. Pedro Russo

Professor of science and society at NOVA University of Lisbon, Portugal

Email: prusso@cienciaviva.pt

Phone: +351 (0) 962 854 775

Relevant Skills

Language proficiency: Indonesian (native), English (CEFR C1 (advanced)).

Programming language: Python (NumPy, Astropy, SciPy, Pandas, Matplotlib, Astroquery, Photutils), R

Softwares: AstroimageJ, IRAF, N.I.N.A., LaTeX, Atlas.ti, Qualtrics